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Portfolio - idealankit.github.io/IdealAnkitPortfolio/



SUMMARY

Aspiring AI/ML and Full-Stack Engineer with experience building **real-time machine learning systems** and **ML-powered web applications**. Strong in **computer vision, deep learning, and backend-frontend integration**, with a focus on turning research ideas into **scalable, production-ready solutions** using **Python, PyTorch, React, and modern web stacks**.

TECHNICAL SKILLS

Machine Learning & AI: Machine Learning, Deep Learning, Computer Vision, CNNs, Metric Learning, Embeddings, Model Evaluation, Threshold Tuning | **Frameworks & Libraries:** PyTorch, OpenCV, Scikit-learn, NumPy, Pandas, Matplotlib, Ultralytics YOLOv8 | **Generative AI (Applied):** LLM fundamentals, prompt engineering, AI pipelines | **Web Development:** HTML, CSS, JavaScript, React | **Programming:** Python, Java, C++, SQL | **Systems & Backend:** Flask, REST APIs, real-time video streaming, logging, performance optimization | **Tools:** Git, GitHub, VS Code, Google Colab | **CS Fundamentals:** DSA, OS, DBMS, OOP

EDUCATION

B.Tech, Computer Science & Engineering | Purnea College of Engineering, Purnea
Nov 2022 – Present | CGPA: 8.64

Expected Graduation: June 2026

Intermediate (PCM + Computer Science) | Kendriya Vidyalaya, Bihta, Patna | 85.6%

April 2020 – March 2022

Matriculation | Shivam Vidya Mandir, Patna | 85.4%

March 2020

EXPERIENCE

- Department of Revenue and Land Reforms – Technical Intern** Dec 2025 – Jan 2026
Supported government IT systems for land and revenue data by assisting in application workflows, data handling, and digital governance processes.
- SmartInternz (AICTE & Salesforce) - Salesforce Developer Intern** Aug 2023 – Sep 2023
Designed CRM workflows and data models using Apex and Lightning components to streamline simulated enterprise business processes.

PROJECTS

- Multi-Face Recognition System (Flagship Project)** | Python, PyTorch, OpenCV, FaceNet, MTCNN | [GitHub Link](#)
 - Implemented a **research-based face recognition system** using **MTCNN** for detection and **FaceNet (InceptionResNetV1)** for **512-D embedding generation**.
 - Designed a **multi-pose enrollment pipeline** to improve generalization and robustness in real-world conditions.
 - Conducted **threshold tuning experiments** to balance false positives and false negatives, achieving **95%+ recognition accuracy**.
 - Built a **real-time inference pipeline** supporting multiple faces per frame at **20–30 FPS with GPU acceleration**.
- Aerial Watch – Drone-Based Intruder Detection System** | Python, OpenCV, YOLOv8, Flask | [GitHub Link](#)
 - Developed a **real-time human detection system** using **YOLOv8**, optimized for streaming video inputs.
 - Implemented **event-driven ML logic** with confidence filtering, cooldown mechanisms, and automated snapshot logging.
 - Integrated ML inference with backend services to support **live video streaming and monitoring dashboards**.
 - Focused on **latency, stability, and continuous inference performance** under real-time workloads.
- Gender-Vision – Real-Time Face & Gender Detection** | Python, OpenCV, CNN, RandomForest | [GitHub Link](#)
 - Built a **real-time computer vision system** combining **CNNs and RandomForest classifiers** for gender prediction.
 - Optimized live video inference and compared classical ML vs deep learning performance.
 - Achieved **~94% accuracy** under real-time conditions.

TRAINING

- IoT Training Program** | iHub DivyaSampark, IIT Roorkee | [Link](#) Mar 2025 – Apr 2025
 - Devised real-time IoT applications using Arduino UNO and NodeMCU, interfacing 5+ sensors.
 - Modeled circuits in Proteus 8, optimizing LED control for 95% accuracy in demos.
- Python for ML, Data Science & Web Dev** | iHub DivyaSampark, IIT Roorkee | [Link](#) Dec 2023 – Jan 2024
 - Executed 5+ ML models and visualizations with Pandas and NumPy in Google Colab, achieving 95% accuracy.
 - Analyzed datasets to support data-driven insights in 40-hour training.

ACHIEVEMENTS & CERTIFICATES

- Secured **Top 1% rank** in **NPTEL Programming in Java**.
- Achieved **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate (Issued by Oracle)**.
- Earned 6 **Microsoft Learn** AI Fundamentals Trophies for AI mastery.
- Completed **NPTEL Cloud Computing**, demonstrating expertise in cloud technologies and services.

ADDITIONAL

- Languages:** English, Hindi
- Interests:** Applied ML research, Technology trends, Production AI, Computer vision systems.